SCHEME OF VALUATION - QP CODE: 25901548

MGU-UGP (HONOURS) REGULAR EXAMINATION-MARCH 2026

FOURTH SEMESTER

MG4DSCSTA203-APPLIED STATISTICAL METHODS

PART A Short Answer Questions

Answer any 7 questions. Each question carries 2 marks

1. Definition/formula of Simple average of price relative – 2 marks.
2. CPI using the family budget expenditure method definition/formula– 2 marks.
   (CPI = $\frac{\Sigma IV}{\Sigma V}$, I = $\frac{P1}{P0}$ x 100 and W = p_0q_0)
3. Key use of a run test is to determine whether a data sequence is random or if it contains systematic patterns – 2 marks.
4. The multiplicative model is given by **Y = T x S x C x I**, T- trend, S- seasonal variation, C- cyclical variation and I – irregular variation – 2 marks.
5. Semi Average method definition – 2 marks.
   (In this method, the data are divided into two equal parts. The average of each part is calculated and plotted to obtain a straight trend line)
6. Definition of Deseasonalisation – 2 marks.
   (Deseasonalisation is the process of removing seasonal effects from time series data. It allows meaningful comparison between different periods.)
7. Any two assumptions of large sample test for mean - 2 marks.
   ($n \geq 30$, randomly selected and independent)
8. One tailed test definition – 2 marks
   (A one-tailed test is a statistical hypothesis that checks for a relationship or difference in one specific direction—either only greater than or only less than a given value)
9. Degrees of freedom in a two-sample t-test = $n_1 + n_2 - 2$ - 2 marks.
10. Null hypothesis for testing the significance of difference between two population means is given by $H_0 : \mu_1 = \mu_2$ - 2 marks.

Part B Short Essay Questions

Answer any 4 questions. Each question carries 6 marks

11. Condition for factor reversal test $I_{ok} \times I_{ko} = \frac{\Sigma pkqk}{\Sigma p0q0}$ - 3 marks.

    Showing Fishers index number satisfies factor reversal test – 3 marks.

12. Explaining time series analysis – 3 marks, example – 3 marks.

13. Explaining moving average method – 6 marks.

14. Any two assumptions – 3 marks, Any two applications – 3 marks

(Independent observations, an independent variable with at least three categorical groups and similar distribution shapes if comparing medians, used to compare three or more independent groups with ordinal or continuous, non-normally distributed data)

15. Null hypothesis H_o : $\sigma^2 = 4$ - 1 mark,
    Alternate hypothesis H_1 : $\sigma^2 \neq 4$ - 1 mark,
    Test statistic $\chi 2 = \frac{(n-1)s^2}{\sigma_0^2}$ = 13.5 – 2 marks
    Conclusion – 2 mark

    (At 5% level of significance, we **fail to reject the null hypothesis**, as the

    calculated chi-square value of 13.5 does not exceed the critical threshold.)

16. Let p be the proportion of male children. We have to test,
    Null hypothesis H_o : $p = 1/2$ - 1 mark,
    Alternate hypothesis H_1 : $p \neq 1/2$ - 1 mark,
    Test statistic $t = \frac{\frac{x}{n} - p_0}{\sqrt{\frac{p_0 q_0}{n}}}$ = 2.896 – 2 marks
    Table value $t_{\propto/2}$ = 1.96 - 1 mark
    Conclusion – 1 mark, reject H_0

Part C Essay Questions

Answer any 1 question. Each question carries 12 marks

17. Table – 3 marks

Commodities	p_0	q_0	p_1	q_1	p_0q_0	p_0q_1	p_1q_0	p_1q_1
Rice	20	28	16	32	560	640	448	512
Wheat	14	18	10	22	252	308	180	220
Pulses	25	20	30	28	500	700	600	840
					1312	1648	1228	1572

    LIN = 93.59756 – 3 marks (formula 2 mark, answer 1 mark)

    PIN = 95.38835 – 3 marks (formula 2 mark, answer 1 mark)

    FIN = 94.48871 – 3 marks (formula 2 mark, answer 1 mark)

18. Null hypothesis H_o : $p_1 = p_2$ - 1 mark,

    Alternate hypothesis H_1 : $p_1 > p_2$ - 1 mark,

    p_1= 0.667-1 mark,p_2 = 0.571 -1 mark,

    Test statistic equation – 2 marks

    Test statistic t =1.23 – 2 marks

    Table value $t_{\propto/2}$ = 1.96 - 1 mark

    Conclusion – 2 mark, no reason to reject H_0